Kyle Wong

Recent Cambridge Master's in astrophysics, applying mathematical-physics and high-performance numerical methods to industry research

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Website: https://kyleyhw.github.io/personal_website/

EDUCATION

Sidney Sussex College, University of Cambridge, Cambridge, England, United Kingdom — 2024 – 2025
MASt/Part III in Astrophysics
Graduated with Good Honours equivalent

Victoria College, University of Toronto, Toronto, Ontario, Canada — 2020 – 2024
HBSc in Physics & Mathematics
Graduated with High Distinction

German Swiss International School, Hong Kong SAR, China — 2010 – 2020
International Baccalaureate (IB) Diploma in 6 subjects

RESEARCH EXPERIENCE

Master of Advanced Study Research Project — Cambridge, England, United Kingdom
Postgraduate Student — October 2024 – September 2025

- Implemented and statistically validated variable cosmological initial conditions in the 21cmSPACE simulation package, enabling efficient forecasting of hydrogen distributions for upcoming radio astronomy experiments including the Square Kilometre Array (SKA).

Canadian Institute for Theoretical Astrophysics
Summer Undergraduate Research Fellowship — Toronto, Ontario, Canada
Undergraduate Researcher — May 2023 – December 2023

- Estimated neutron star tidal deformability from LIGO gravitational wave data using high-dimensional MCMC parameter estimation, and contributed new equation-of-state models to the collaboration's analysis pipeline.

McGill Space Institute Summer Undergraduate Research Award — Montreal, Quebec, Canada
Undergraduate Researcher — May 2022 – April 2023

- Explored integration of statistical priors into the power spectrum estimator for the Hydrogen Epoch of Reionization Array (HERA) collaboration's cosmic dawn experiment data pipeline, improving signal inference from radio astronomy observations.

FEATURED PERSONAL PROJECTS

mercury_strategies [Private Repository] (work in progress)

- Developed an event-driven engine for capturing alpha in weather-derivative prediction markets (Polymarket), managing a dockerized infrastructure on AWS with integrated risk management. Features a provider-agnostic collection and analysis layer, asynchronous watcher architecture for real-time signal generation, and automated execution with velocity rotation, local orderbook tracking, and risk-aware position sizing via the Kelly Criterion. Includes 4 months of manual prediction market trading to gain market knowledge, intuition, and experience.

sound_simulation (https://github.com/kyleyhw/sound_simulation)

- Engineered an N-dimensional Finite-Difference Time-Domain (FDTD) simulator in Python that solves the acoustic wave equation via an explicit leap-frog scheme on a Cartesian grid, with CFL-enforced stability, hard-wall boundary conditions, arbitrary driver waveforms, and sensor arrays. Accelerated the 2D stencil kernel with a fused numba-JIT pass using prange parallelism for a roughly 65× speedup over the SciPy baseline.

agent_evolve (https://github.com/kyleyhw/agent-evolve)

- Created a multi-model, cooperative evolutionary code-optimisation agent skill. Supervisor, explorer, and reviewer roles can each be filled by any LLM through CLI, with candidates evolving on isolated git branches. Features interactive visualization and surfaces the winner as a pull request for human approval.

HONOURS, AWARDS AND SCHOLARSHIPS

- Dean's List Scholar in the Faculty of Arts & Science — 2022, 2023, 2024
- Canadian Institute for Theoretical Astrophysics
  Summer Undergraduate Research Fellowship (CITA SURF) — 9,500 CAD, 2023
- Birkenshaw Family Scholarship II — 1,000 CAD, 2023
- McGill Space Institute (now Trottier Space Institute)
  Summer Undergraduate Research Award (MSI SURA) — 7,000 CAD, 2022
- David and Louise Fraser Scholarship — 2,500 CAD, 2022
- University of Toronto Scholar — 1,500 CAD, 2022
- Birkenshaw Family Scholarship — 1,000 CAD, 2022

SELECTED COURSES

- Gravitational Waves & Numerical Relativity (Cambridge, 2025)
- Astrostatistics (Cambridge, 2025)
- Cosmology (Cambridge, 2024)
- General Relativity (Cambridge, 2024)
- General Relativity (Toronto, 2024)
- Relativity Theory II (Toronto, 2024)
- Relativity Theory I (Toronto, 2023)
- Quantum Mechanics II (Toronto, 2023)
- Computational Physics (Toronto, 2023)
- Computational Astrophysics (Toronto, 2023)
- Advanced Classical Mechanics (Toronto, 2023)
- Geometry of Curves and Surfaces (Toronto, 2023)
- Partial Differential Equations (Toronto, 2023)

SKILLS

- **Technical:** Scientific computing, machine learning, statistical inference, simulation, time-series analysis, data visualization, and high-performance computing, primarily in Python within Linux environments.
- **AI Tools:** Claude Code, Google Gemini, Gemini CLI, ChatGPT, and Google Antigravity for automated software engineering, academic and scientific research, and coding.

LANGUAGES

- English (fluent, language of education)
- Cantonese (fluent, mother tongue)
- Mandarin (formally learned 10 years)
- German (formally learned 10 years)

RECENT MEMBERSHIPS

- Sidney Sussex College Football Team, Premier Division
- Institute of Astronomy Football Club
- Cambridge University Mountaineering Club
- Cambridge University Table Tennis Club
- Cambridge University Hong Kong Postgraduate Student Association
- Cambridge University Chinese Society

PERSONAL INTERESTS

- Rock climbing (9+ years experience both indoors and outdoors, with awards won at amateur, inter-high school and inter-university competitions. Also led a rock climbing extra-curricular activity during high school, including captaining the competition team)
- Soccer (7+ years league participation)
- Scuba diving (PADI open water certified)
- Travelling